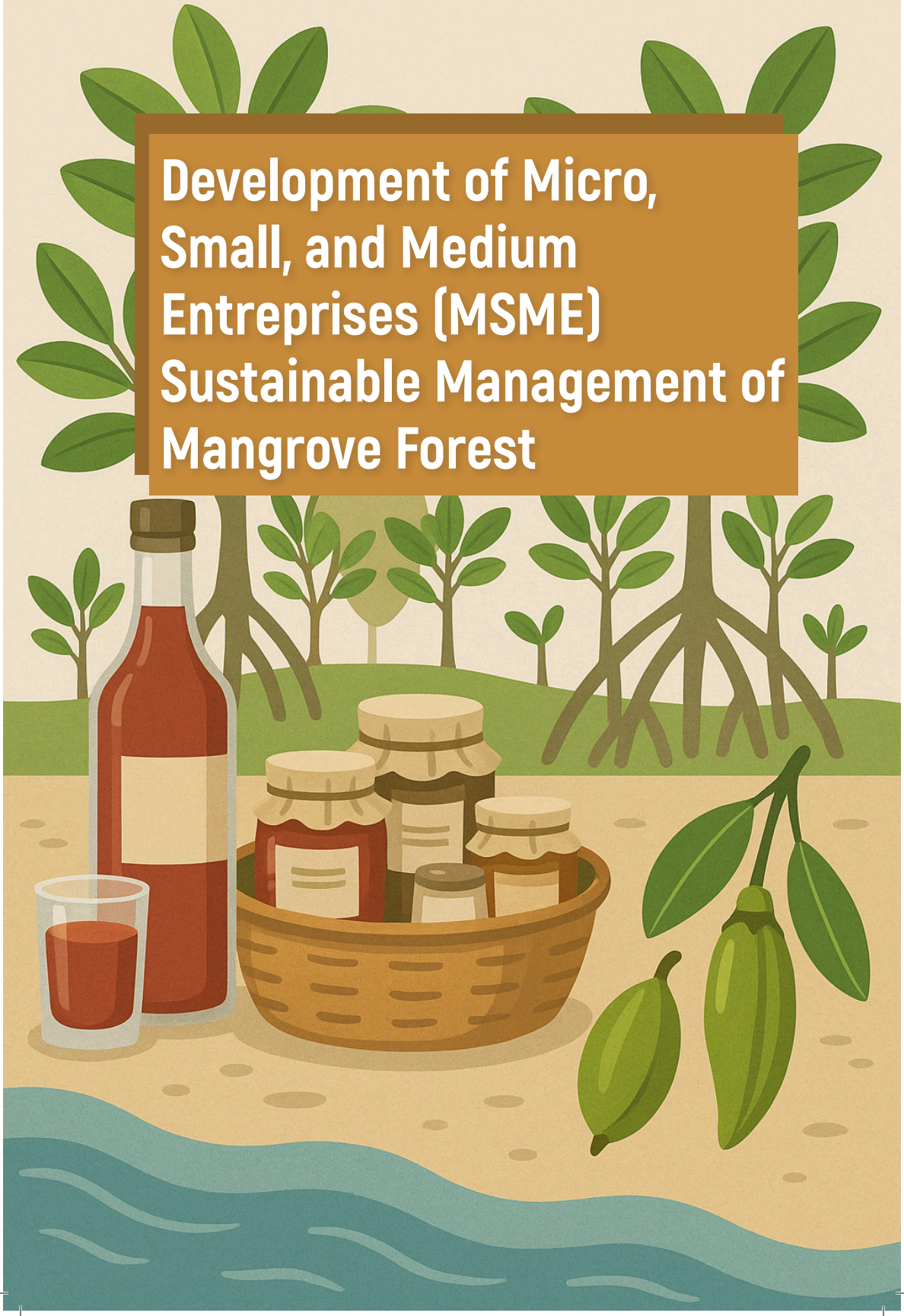
Development of Micro, Small, and Medium Entreprises (MSME) Sustainable Management of Mangrove Forest



Development of Micro, Small, and Medium Entreprises (MSME) Sustainable Management of Mangrove Forest

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FOREWORD

Mangrove ecosystems are a critical component of the ASEAN region's coastal landscapes, delivering vital ecological, economic, and social services. They provide habitats for diverse species, protect shorelines from erosion, buffer communities against storms, sustain fisheries and livelihoods, and serve as one of the world's most important blue carbon ecosystems. With an estimated 4.5 to 5 million hectares, approximately 30 percent of the global mangrove area, Southeast Asia holds the largest extent of mangroves worldwide. Their conservation and sustainable management are therefore of both regional and global significance. Yet, these ecosystems continue to face pressures from unsustainable exploitation, land conversion, pollution, and climate change, underscoring the urgent need for coordinated and science-based management.

Within ASEAN, Indonesia and other Member States host some of the largest mangrove areas in the world, yet these ecosystems are under increasing pressure from conversion, overuse, and climate change impacts. While mangroves provide invaluable environmental services, including habitats for marine biodiversity, coastal protection, carbon storage, and raw materials for local livelihoods, they are also central to the well-being of coastal communities.

This guideline has been developed to provide direction on how micro, small, and medium enterprises (MSMEs) and cooperatives can play a stronger role in sustainable mangrove management. By linking mangrove ecosystem services with community-based businesses, the guideline aims to support inclusive economic growth, reduce disparities, and build social resilience. It presents

practical steps for identifying opportunities, developing MSMEs, and strengthening cooperative governance while ensuring that mangrove ecosystems remain protected and sustainably used.

Importantly, this publication supports the ASEAN Strategy on Mangrove Ecosystem Management, particularly Strategy 5: Enabling understanding among AMS in facing common, regional, and global challenges related to mangrove ecosystem management. Through knowledge-sharing, peer learning, and regional cooperation, this guideline encourages ASEAN Member States to integrate community empowerment, cooperative development, and ecosystem sustainability in addressing shared challenges.

We hope this guideline will serve as a valuable reference for governments, local communities, NGOs, academia, and the private sector, inspiring collaboration and innovation in harnessing the economic potential of mangroves while safeguarding them for future generations.

Author

Indonesia, 2025

I.

INTRODUCTION

A. Background

The mangrove ecosystem is a unique ecosystem found in coastal areas. The area of mangroves in the world is 16.53 million hectares. This was reported by Tresiana et al. (2022) based on data from the World Atlas of Mangroves. Of this area, ASEAN contributed 33.5% or 5.54 million hectares. Meanwhile, Indonesia's mangrove area is 3.49 million hectares, covering 19% of the world's total and 63% of ASEAN's mangroves. The potential area of Indonesia's mangroves is currently facing the problem of damage caused by various factors, which causes damage in various locations. Data from research in Indonesia shows that the area of Indonesia's mangroves decreased by 3.75 million hectares in 2010. In the past five years, 260,859.32 hectares of mangroves have been degraded. Finally, in 2020, mapping showed a total mangrove area of

3,311,207.45 hectares, with non-critical conditions covering an area of 2,673,583.14 Ha (80.74%) and critical mangroves covering an area of 637,624.31 Ha (19.26%). According to the results of mangrove mapping carried out by the Directorate of Soil and Water Conservation of the Ministry of Environment and Forestry (2021), the mangrove area is 3,364,080 Ha (Itfan and Soetjipto, 2024). The remaining mangrove forests must be managed professionally, including protection, preservation, and sustainable use.

The biological and ecological conditions of mangrove forests are very unique and have the potential for extensive environmental services. These environmental services include various benefits for humans and the environment, such as habitats and breeding grounds for marine biota, coastal abrasion barriers, carbon absorbers, and sources of food and industrial raw materials. Various feasible business activities that are starting to be carried out by village communities are ecotourism, traditional medicine, and silvofishery. Researchers and entrepreneurs have also begun to study the use of carbon stored in the mangrove ecosystem.

Economic growth in Indonesia in the last two decades has experienced a growth of 5%. However, this economic growth is not accompanied by an even distribution of the economy in Indonesia. On the contrary, this economic growth is accompanied by disparities in society (Suroto, 2023). This emerging disparity can be handled by economic activities that take place in various sectors of all stakeholders, namely: government, and society, one of which is mangrove forest business activities. Mangrove forest business activities that are starting to grow and develop in Indonesia must receive government attention and support so that they become a sustainable social and economic force for society. The growth and development of this community's

economic strength requires support from the management capacity of village community cooperatives, especially in preparing and developing human resources, governance, and capital.

Therefore, guidelines are needed in accordance with the business fields that are focused on the UMKM forum, as well as building community participation in the framework of sustainable mangrove forest management. This guideline was developed as part of an action strategy 4 number 5 to Involving and enhancing capacity of local community including local champions, traditional figures, and village leaders as well as women and youth in the planning, implementation, monitoring, and evaluation of the program/activities, through facilitating community discussions, field school, information and education center and platform, and sharing learning and spirit among communities, with the presence of the local leadership.

B. Objective

This guideline is designed to identify the steps for developing MSMEs through cooperatives to produce a sustainable mangrove ecosystem by considering the correlation between community activities and the mangrove ecosystem in order to maximize the potential for community development through MSME activities that can increase community income and ensure the sustainability of the mangrove ecosystem.

C. Scope

The guidelines will discuss in detail: 1) Characteristics and uniqueness of mangrove forests; 2) Environmental services of mangrove ecosystems; 3) Management and sustainable development of mangrove ecosystems; 4) Cooperatives

and MSMEs; 5) MSME development; and 6) Stages of MSME development. This is in line with the five strategies of AMNET (ASEAN Mangrove Network), namely: 1) strengthening regional cooperation, 2) harmonization of policies and regulations, 3) capacity building and education, 4) science-based research and monitoring, 5) community empowerment and community-based initiatives.

D. Guideline Targets

The goal of this guideline is to achieve the human happiness index (sustainable development), namely a development process based on the principle of "meeting the needs of the present without compromising the needs of future generations" (Brundtland Report from the UN, 1987). Sustainable Development Goals (SDGs) are a global action plan agreed upon by world leaders, including Indonesia, to end poverty, reduce inequality, and protect the environment. Sustainable Development Goals (SDGs) contain 17 Goals and 169 Targets that are expected to be achieved by 2030 (Alikodra, 2020). The guidelines are made to be understood and implemented by all stakeholders, namely: the government, NGOs, academics, philanthropic industry, local communities, and mass media.

E. Glossary

- Micro, Small, and Medium Enterprises (MSME)

  Micro, Small, and Medium Enterprises (MSMEs) are a grouping of businesses based on revenue and production output, marketed to the public, managed by individuals or business entities. In this guideline, these businesses are

defined as those utilizing mangrove forests to produce value-added products that can be marketed by the public on an MSME scale.

- Sustainable Management

  Sustainable management in these guidelines is defined as a step in utilizing mangrove potential while maintaining social, ecological, and environmental aspects so that communities surrounding the mangrove ecosystem can continue to utilize it without damaging the mangrove forest and maintaining its sustainability. The guidelines present several alternative solutions in the form of activities that communities can undertake to implement the concept of sustainable management in mangrove forests.

- Mangrove Forest

  Mangrove forests are complex ecosystems composed of diverse plant species that form the transition between land and sea. Typically found in brackish water areas, they are salt-tolerant, making them a unique coastal ecosystem and a valuable natural resource, supporting the diverse flora and fauna that inhabit them.

II.

MANGROVE FORESTS: ECOSYSTEM SERVICES AND SUSTAINABLE MANAGEMENT

Mangroves are characteristic of coastal vegetation, estuaries, or river mouths, and provide shelter for various types of fish, shrimp, crabs, birds, and primates in tropical and subtropical areas. Mangrove forests consist of individuals of various types of plant communities in tidal coastal waters, often called mangrove forests or brackish water forests. Most of the vegetation is dominated by mangrove species, so they are called mangrove forests and are also called brackish water forests because the forests grow on land that is always inundated by brackish water. The meaning of mangrove in plant ecology is used for shrubs and trees that grow in shallow intertidal and subtidal areas in tropical and subtropical tidal swamp ecosystems (Choong et al., 1990).

Mangrove plants are unique and distinctive because they are a combination of the characteristics of plants that live in coastal areas. Generally, mangroves have a prominent root system called respiratory roots (pneumatophores) (Choong et al., 1990). The mangrove forest ecosystem is composed of trees and shrubs, has tolerance to salt, is one of the unique coastal ecosystems, and is a very valuable natural resource to support the existence of the diversity of flora and fauna that live in it. The mangrove forest ecosystem covers only 2% of the earth's surface, making it one of the rare ecosystems in the world (Eddy et al., 2017).

The uniqueness of the Indonesian mangrove ecosystem is that it has the highest species diversity. Several types of mangroves that are often found in Indonesia are grouped into several families, namely, the Rhizophoraceae family, the Avicenniaceae family, the Sonneratiaceae family, the Verbeneaceae family, and the Meliaceae family (Rahman and Pane, 2000). However, in the last two to three decades, almost 50% of the total area of mangroves in Indonesia has been lost, from around 6.7 million ha to around 3.2 million ha. Java and Bali are the islands with the highest damage, which is around 88%. Previously, these two islands had around 171,500 ha, but now only around 19,577 ha (Eddy et al., 2017).

The decreasing distribution of mangrove forests is due to human activities (anthropogenic), which contribute the most to the destruction of mangrove forests in Indonesia. Conversion of mangrove forests for fisheries, plantations, agriculture, salt ponds, settlements, industry, agriculture, logging, and mining activities are the main anthropogenic activities causing the degradation and loss of mangrove forests in Indonesia (Eddy et al., 2017). Other activities, such as aquaculture, and infrastructure, namely ports, industry, trade centers, and housing, are the causes of the reduction in mangrove resources (Purnobasuki, 2011).

A. Mangrove Ecosystem Environmental Services

Mangrove environmental services are very broad, covering various benefits for humans, the environment, and development. Environmental services are closely related to the role/function of mangrove forests, such as habitats and breeding grounds for marine biota, including various types of wildlife (birds, primates, snakes, and monitor lizards), coastal abrasion barriers, carbon absorbers, and sources of food and industrial raw materials. Mangroves also have socio-cultural and economic services, such as recreation, tourism, and education.

Mangrove environmental services are very broad and complex, according to Alikodra (2020), and can be divided into:

1. Provisioning Services: Raw materials for food, industry, and fresh water, because mangrove forests play a role in maintaining the balance of fresh water in coastal areas.

2. Regulating Services: Able to withstand abrasion, wave dampers, carbon absorption (helps reduce the greenhouse effect, and climate regulation), regulates the local climate by dampening wind and maintaining humidity.

3. Supporting Services: Habitat and breeding grounds and nursery grounds for various types of fish, shrimp, crabs, and various types of marine biota, and formation of coastal soil.

4. Cultural Services including beliefs: Recreation and ecotourism, education, research, and education, as well as the greatness of the creator, deep ecology/ecosophy.

5. Aesthetics: Mangrove forests have aesthetic value that can improve the quality of life of the community.

Given the importance of mangrove environmental services, conservation needs to be implemented, including protection, planting, restoration, and sustainable use. These activities include preventing deforestation, planting mangroves in damaged areas, and the wise management of mangrove resources.

B. Sustainable Utilization and Management

Sustainable management is the utilization of mangrove land and forest products. This activity includes efforts to preserve and prevent mangrove forest damage (Davinsy et al., 2015). Sustainable mangrove forest management is highly effective for carbon sequestration and storage. Another function of this management is to ensure that mangrove forests are not damaged, thus preventing abrasion, absorbing emissions, and maintaining their biological and ecological functions.

Specifically, mangrove forests are considered capable of storing four times more carbon reserves than other tropical forests. In addition, mangrove forests also have a five times faster absorption rate of carbon elements in the air compared to forests on land (Rifandi, 2020). Therefore, overall, in addition to the potential to reduce negative impacts on the environment, sustainable mangrove forest management can also improve the economy of the surrounding community through the development of their businesses within the framework of MSMEs.

Of course, the management of mangrove forests must pay attention to the environmental conservation aspect. This aspect is important to note because it is necessary to protect biodiversity, especially flora and fauna that make mangroves a habitat, and ensure that the environment is maintained for the sake of the sustainability of the ecosystem and various lives. Furthermore, it is necessary to carry out climate change

mitigation, which will later have an impact on the environment around the mangrove forest, and of course, at the same time, can help anticipate various disasters that will occur.

The development of mangrove UMKM (for example, Sustainable Mangrove UMKM) includes silvofishery, carbon, medicinal plants, and food businesses, which can increase conservation efforts. The capacity of MSME actors can be fostered so that they develop into professional activities as part of sustainable mangrove forest management. The use of mangrove forest resources must be sustainable and not damage the environment. This aspect certainly needs to be considered because it is closely related to the activities of communities and MSMEs around the coast.

Concerning cultivation activities, sustainable management is needed in the creation of shrimp and fish ponds that must maintain the mangrove forest ecosystem. In addition, it is also important to empower communities within the framework of MSMEs by developing knowledge of mangrove and fisheries business management for the community to obtain optimal social, economic, and ecological benefits from the mangrove forest ecosystem.

Specifically, in the framework of MSMEs, sustainable mangrove utilization can be carried out through the following activities.

1. **Ecotourism Development**

   Mangrove ecotourism is one of the efforts to preserve mangrove forests. The existence of good and healthy mangroves in coastal areas can increase the resilience of coastal communities to climate change and reduce the risk of natural disasters such as storms, tsunamis, and coastal abrasion. In addition to being able to produce products of economic value, the mangrove forest ecosystem can also

generate income for the community through ecotourism activities. The existence of mangrove forest ecotourism will certainly make the wheels of the economy for MSMEs run well.

Every MSME actor needs to be given an understanding to participate in managing mangrove forests. This has been implemented quite a lot in various regions in Indonesia, but of course, this aspect of ecotourism development must be improved with good promotion and access, provision of adequate infrastructure, providing education and socialization to visitors, and ensuring that the community around the mangrove forest area has received education related to ecotourism activities.

2. **Carbon Trading**

Carbon trading is a system where permits to produce greenhouse gas (GHG) emissions are traded between countries or as business entities. The main goal is to reduce GHG emissions globally through economic incentives for parties who successfully reduce emissions, or in other words, carbon trading. The carbon trading mechanism involves buying and selling carbon credits or emission quotas. Carbon credits represent the right to produce a certain amount of GHG emissions (usually expressed in tons of CO_2).

It is an incentive for companies or countries that exceed their emission quotas to buy carbon credits from entities that successfully reduce their emissions. This creates an incentive for entities to reduce emissions and generate carbon credits that can be traded. The goal is to reduce GHG emissions, especially CO2, and help countries achieve their emission reduction targets. It is done through a carbon exchange, which is a platform for carbon credit transactions

3. Waste and Garbage Management

The development of effective and efficient waste and garbage management is certainly needed to ensure that the mangrove forest ecosystem can be maintained properly. The types of waste and garbage that are often found in mangrove forests include household waste, such as plastic, bottles, bags, and packaging, as well as organic waste such as food scraps. In addition, other waste that is also often found includes cork, broken glass, and oil spill tar balls, as well as industrial waste. There needs to be a 3R effort (Reduce, Reuse, Recycle) in the development of MSMEs in mangrove forest management. In addition, it is necessary to pay attention to appropriate technology in supporting mangrove forest management activities. Plastic waste that cannot be broken down, including plastic bottles, is often caught in the mangrove roots and must be removed manually.

Ecobricks are also a popular method for processing plastic waste into building materials. Plastic waste recycling is the process of reprocessing plastic waste into new, useful products, with the aim of reducing environmental impacts and reusing plastic waste. This process involves several stages, such as collecting, sorting, cleaning, shredding, and washing. The processed plastic can be reused in making new products, such as clothes, shoes, bags, or furniture.

Currently, waste banks have developed, which are waste management systems that are run like banks, where people can save dry waste that has economic value (such as plastic, paper, and metal) and get rewards in the form of money. Residents who save waste are called customers and have savings books like regular bank customers.

4. **Bioprospection**

Bioprospection is systematic search, classification, and investigation of biological natural resources for commercial purposes. This includes the search for new chemical compounds, genes, proteins, and other products that have economic value and potential in various fields such as health, agriculture, and cosmetics. Bioprospection aims to find and identify biological resources that can be used commercially, both those that are already known and those that are not. Various research institutions (universities, plantations, agriculture, and BRIN) have started various studies of plant and microbial types. The potential of this bioprospection can be a very useful activity in MSME activities that will support the economy of the community, especially around the coast, which has great potential to be developed.

5. **Mangrove Silvofishrey**

Silvofishery is a fishery cultivation system that combines mangrove forests with fisheries cultivation, such as shrimp, fish, crab, or shellfish cultivation. This system allows the preservation of mangrove forests while producing fishery commodities, thus becoming a sustainable solution for shrimp farming and fisheries cultivation in coastal areas. Silvofishery in the mangrove ecosystem has enormous potential to be developed for MSMEs around the coast. Several marine biota and coastal plant cultivations that can be managed by the community will become a new source of livelihood that can increase community income.

6. **Food and Beverage Industry**

The food and beverage industry from mangrove resources has begun to develop in Indonesia. This activity offers great potential, with various existing products and opportunities for further innovation. These products include processed fruit such as syrup, jam, sticky toffee dessert, and flour, as well as processed leaves such as chips, sticks, and crackers. The use of mangroves also provides social and economic contributions to coastal communities, and of course, it is hoped that this activity can also help maintain the sustainability of the mangrove ecosystem. MSME activities around the mangrove ecosystem can be a source of raw materials for industrial activities, as a source of raw materials for production.

III.

DEVELOPMENT OF MICRO, SMALL, AND MEDIUM ENTERPRISES (MSME)

Cooperatives are business entities whose members are individuals or cooperative legal entities whose business activities are based on cooperative principles and family principles that aim to improve the welfare of members and the community. Cooperatives are a people's economic movement based on cooperative values such as freedom of entry and exit, democratic management, and fair distribution of business results (SHU).

The meaning of cooperative Small and Medium Enterprises, micro-enterprises, is cooperation, which is one of the human instincts that needs to be continuously developed. In economic life, the prosperity and progress of all people can be achieved. Cooperation in cooperatives aims to

improve the welfare of members through businesses managed by members, as owners and at the same time users of services. Although in the field of economic business it is necessary to consider the principles of efficiency, the human factor and human values in cooperatives must be upheld. Cooperatives as a cooperation organization of members whose role is very important because they are a business structure that follows economic democracy. This cooperative can strengthen social infrastructure and distribute assets that can eliminate the gaps in society (excluding land and buildings) and annual (Suroto, 2023).

To achieve a more humane, just, and democratic economic life system, the common people must be invited and allowed to improve their living status through cooperatives, so that economic power and resources can be owned, managed, and utilized more evenly to avoid the centralization of economic power in certain groups. In order for the community to be able to use cooperatives as a means of struggle and a forum for economic cooperation, educational activities need to be continuously developed. Thus, the weak social groups will gradually have attitudes, behaviors, and values that are more suitable for their progress and the progress of society as a whole.

The cooperative principle is a system of abstract ideas that are guidelines for building effective and durable cooperatives. The latest cooperative principles developed by the International Cooperative Alliance (International Federation of Non-Governmental Cooperatives) are: open and voluntary membership; democratic management; member participation in the economy; freedom and autonomy; and development of education, training, and information. According to Law no. 25 of 1992 concerning cooperatives, the cooperative principles are: voluntary and open membership; management is carried out

democratically; distribution of SHU is carried out fairly according to the business services of each member; limited remuneration for capital; independence; cooperative education; cooperation between cooperatives.

Micro, Small, and Medium Enterprises (MSMEs) is an abbreviation of micro, small, and medium enterprises, which are productive businesses owned by individuals or business entities, and meet the criteria as micro, small, and medium enterprises. Following Law No. 20 of 2008 concerning Micro, Small, and Medium Enterprises, micro-enterprises are businesses with a maximum net worth of IDR 50 million (excluding land and buildings), and annual sales of up to IDR 300 million. Small businesses, businesses with a net worth of more than IDR 50 million to IDR 500 million (excluding land and buildings) and annual sales of more than IDR 300 million to IDR 2.5 billion. Medium businesses are businesses with a net worth of more than IDR 500 million to IDR 10 billion (excluding land and buildings) and annual sales of more than IDR 2.5 billion to IDR 50 billion.

Micro, small, and medium enterprises, also known as small and medium enterprises, are types of companies in Indonesia owned by individuals or business entities in accordance with the criteria set by Law No. 20 of 2008 concerning Micro, Small, and Medium Enterprises. MSMEs play an important role in the Indonesian economy, both in creating jobs, driving economic growth, and improving community welfare.

The objectives of developing MSMEs in the context of sustainable and continuous mangrove forest management are as follows:

- Improving the welfare of local communities through the development of MSMEs;

- Increasing the income of local communities through sustainable mangrove forest management activities.
- Achieving sustainable and sustainable mangrove forest conditions.

Meanwhile, the benefits of developing MSMEs for sustainable and continuous mangrove forest management are:

- Increasing awareness and understanding of the importance of sustainable mangrove forest management.
- Increasing the capability of mangrove forest management MSMEs to face challenges and opportunities.
- Increasing the performance and competitiveness of mangrove forest management MSMEs.

A. Developing MSMEs

The goal of improving community welfare and creating jobs is to determine the aspect of success in mangrove forest management. The success in question is that MSMEs and surrounding communities that successfully manage mangrove forests will certainly increase the income of the surrounding community so that they can have better access to education and health, create good jobs, and, of course, be supported by better infrastructure.

Training programs and community awareness are very important to support this aspect so that the community can improve their quality of life and ensure the sustainability of the mangrove ecosystem. The community's capacity must be increased to participate in mangrove rehabilitation and management, including in mangrove-based economic activities with optimal governance.

Micro, Small, and Medium Enterprises (MSMEs) are a grouping of types of businesses based on the criteria of types of activities, assets, and turnover obtained. The criteria for the grouping that was previously carried out were based on net assets and annual sales results, but these criteria can be changed and refined through the Job Creation Law, namely, the criteria used are based on business capital or annual sales results (Muttaqien et al., 2022).

Micro, Small, and Medium Enterprises (MSMEs) in Indonesia currently have a fairly important role, especially in the country's economy. The country's unstable economy encourages the government to focus on empowering MSMEs. In addition to running the country's economy, MSMEs also provide opportunities to absorb a large workforce and provide opportunities for MSMEs to develop and compete with companies or industries (Sedyastuti, 2018).

However, the important role of MSMEs i the economy above has quite a big challenge because of the many obstacles faced, such as limited infrastructure, complex licensing access, bureaucracy, corruption, and high levels of levies. Although MSMEs can run quite stably in the midst of the global crisis, MSMEs are faced with challenges that come from within the country, namely labor wages, illegal levies, and so on. Therefore, it is necessary to strengthen capacity, especially governance, quality of human resources, capital, and clear mechanisms and regulations as a guarantee for the sustainability of good MSMEs and can be a buffer for the national economy (Sedyastuti, 2018).

B. Stages of MSME Development

Some steps that must be taken in developing mangrove forest MSMEs include:

1. Planning
    a. Identification and socialization to stakeholders: Conducting data collection and coordinating with stakeholders using tools such as questionnaires and infographics or distributing socio-economic data around the mangrove ecosystem. This activity can be done through interviews, FGDs, and surveys.
    b. Identification of actors and potential MSMEs around the mangrove forest: This can be done by mapping natural resources and local economic activities around the mangrove ecosystem through participatory observation, surveys, and Strengths, Weaknesses, Opportunities, Threats (SWOT) analysis.
    c. Preparation of MSME development plans: Interpreting the identification results and formulating development steps.
2. Implementation
    a. Community socialization and training: Preparing training materials for both theory and practice and appointing facilitators or assistants with a structured and sustainable flow of activities.
    b. Infrastructure development and provision of supporting facilities: Preparing a budget and financing for MSME development activities based on the results of monitoring and observation of needs in the field.

c. Support for MSMEs: MSMEs operating around mangrove forests can be provided with mentoring facilities and partnership channels from various sectors.

3. Monitoring

a. Observation of biodiversity in the mangrove forest ecosystem: Conducting an inventory of flora and fauna in the mangrove ecosystem and analyzing their relationship and impact on MSME activities. This activity can be carried out with transects and sampling blocks as well as basic ecological analysis in the form of diversity indices and direct interviews with related MSMEs.

b. Monitoring of MSMEs operating around the Area: Conducting an assessment of participation and potential sustainability of MSME activities by conducting assessments and interviews, as well as field surveys;

c. Data collection and preparation of monitoring report: Collecting all identification results, interviews, surveys, mapping, and other data for further analysis both quantitatively and descriptively and conductingvarious literature studies as a form of validation and verification of the results.

4. Evaluation

a. Analysis of the achievement of MSME targets for sustainable mangrove forest management.

b. Evaluation of the environmental and socio-economic impacts of the community and MSMEs around the mangrove forest area and Identification of obstacles and solutions for MSMEs, to ensure the implementation of MSME governance.

c. Preparation of the final evaluation report and review and revision of the MSME plan for sustainable mangrove forest management, so that professional business strength is obtained.

IV.

CLOSING

Various potential commercial ventures from mangrove resources continue to be developed professionally. Micro, Small, and Medium Enterprises (MSMEs) around mangrove forests can utilize various natural resources through ecotourism activities, silvofishery, medicines, and resources to create products of economic value, such as food, drinks, and crafts. Through socio-economic improvement, the community will maintain the existence of mangrove forests. Through MSMEs around mangrove forests, it is hoped that various economic benefits can be increased, such as increasing community income, opening up employment opportunities, and developing tourism potential.

REFERENCES

Alikodra, HA. (2020). Era Baru Konservasi SDAL: Membumikan Ekosofi bagi Keberlanjutan Umat. IPB Press. Bogor.

Choong, E. T., Wirakusumah, R. S., & Achmadi, S. S. (1990). Mangrove forest resources in Indonesia. *Forest Ecology and Management*, *33*, 45-57. Eddy, S., Mulyana, A., Ridho, M. R., & Iskandar, I. (2017). Dampak aktivitas antropogenik terhadap degradasi hutan mangrove di Indonesia.

Davinsy, R., Kustanti, A., & Hilmanto, R. (2015). Kajian pengelolaan hutan mangrove di Desa Pulau Pahawang Kecamatan Marga Punduh Kabupaten Pesawaran. *Jurnal Sylva Lestari*, *3*(3), 95-106.

Direktorat Konservasi Tanah dan Air. (2021). Peta mangrove nasional. Jakarta: Direktorat Konservasi Tanah dan Air, Ditjen PDASRH.

Eddy, S., Mulyana, A., Ridho, M. R., & Iskandar, I. (2017). Dampak aktivitas antropogenik terhadap degradasi hutan mangrove di Indonesia.

Itfan, I., & Soetjipto, W. (2024). Dampak Ketergantungan Sosial Ekonomi Desa Terhadap Luasan Hutan Mangrove di Indonesia. *Jurnal Ekonomi dan Kebijakan Publik*, *14*(1), 61-75.

Muttaqien, F., Cahyaningati, R., Rizki, V. L., & Abrori, I. (2022). Pembukuan Sederhana Bagi UMKM. *Indonesia Berdaya*, *3*(3), 671-680.

Purnobasuki, H. (2011). Ancaman terhadap hutan mangrove di Indonesia dan langkah strategis pencegahannya. *Bulletin PSL Universitas Surabaya*, *25*(2011), 3-6.

Rahman, A., & Pane, E. (2000). Pengaruh Jarak Tanam Beberapa Jenis Tanaman Mangrove Terhadap Pertumbuhan Vegetatif di Lokasi Tanah Timbul Kecamatan Bandar Khalipah Kabupaten Deli Serdang Propinsi Sumatera Utara.

Rifandi, R. A. (2020). Estimasi stok karbon dan serapan karbon pada tegakan pohon mangrove di Hutan Mangrove Trimulyo, Genuk, Semarang. *Journal of Environmental Science Sustainable*, *1*(2), 11-18.

Suroto. (2023). Koperasi lawan tanding kapitalisme.

Sedyastuti, K. (2018). Analisis pemberdayaan UMKM dan peningkatan daya saing dalam kancah pasar global. *INOBIS: Jurnal Inovasi Bisnis Dan Manajemen Indonesia*, *2*(1), 117-127.

Tresiana, N., Duadji, N., Febryano, I. G., & Zenitha, S. A. (2022). Saving mangrove forest extinction in urban areas: will government interventions help. *International Journal of Sustainable Development & Planning*, *17*(2).